TITLE PAGE

- **Problem Statement ID – Question 5A**
- **Problem StatementTitle-Grade Change Intiligence in Paper Making Process**
- **Theme- Industry 4.0 – AI for Smart Manufacturing**
- **PS Category- Software**
- **Student Name – Atharv Vijaykumar Patil**
- **Student ID – 23BSA10028**
- **Site Live URL-** https://honeywellgradechangeintligence-stejkkck6udc6tkmh9v8tq.streamlit.app/
- **GitHubrepo:**https://github.com/AtharvCodeCraft/HoneywellGradeChangeIntligence
- **ZipFile :** https://drive.google.com/file/d/1Pqk7IARvT0Fv9ClXjWlUfx5YqhdepoZX/view?usp=sharing

Grade Change Intelligence: AI-Powered Decision Support for Paper Manufacturing

Proposed Solution:

Detailed explanation of the proposed solution:

- AI-powered system predicts paper grade-change risk using XGBoost.
- SHAP Explainable AI identifies the key process parameters influencing predictions.
- Smart dashboard provides recommendations, optimal setpoints, and downloadable PDF reports.

How it Addresses the Problem:

- Detects potential quality issues early to reduce production losses and downtime.
- Assists operators with data-driven recommendations for faster decision-making.
- Improves product quality and process efficiency through continuous monitoring.

Innovation & Uniqueness:

- Combines Machine Learning with Explainable AI for transparent predictions.
- Integrates AI recommendations, correlation analysis, and optimal setpoint suggestions.
- Enables operator feedback and automated PDF reporting in a single industrial dashboard.

TECHNICAL APPROACH

Technical Approach:

Development and Visualization: Python, Streamlit, Pandas, NumPy, Matplotlib

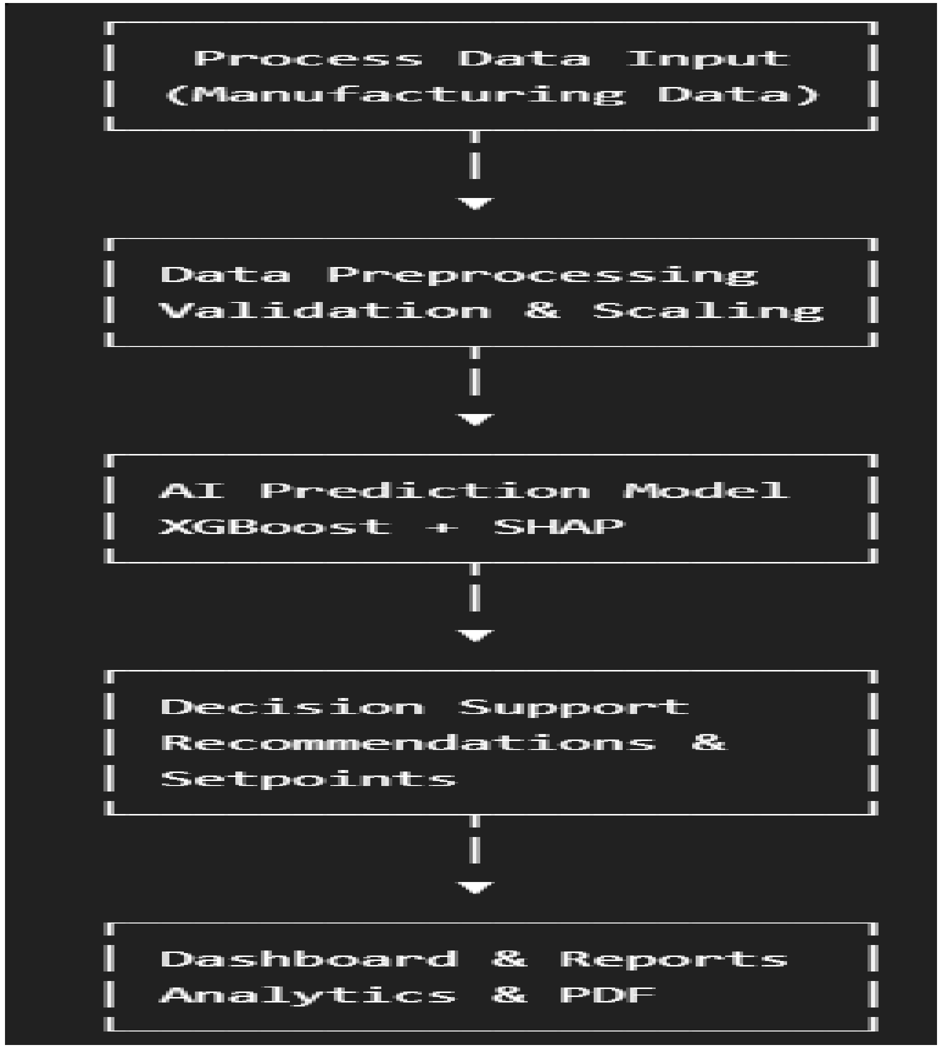
Machine Learning: XGBoost, SHAP, Scikit-learn, Joblib

Report Generation : ReportLab & GitHub

Working Prototype:

- Interactive Streamlit dashboard with real-time process analysis.
- AI-generated recommendations, correlation heatmap, and confidence analysis.
- PDF report generation with operator feedback support.

Methodology and process for implementation:



FEASIBILITY AND VIABILITY

Analysis of the Feasibility of the Idea

- Uses proven technologies (Python, XGBoost, SHAP, Streamlit) that are reliable and industry-ready.
- Easily integrates with existing manufacturing data and process monitoring systems.
- Scalable solution that can be extended to multiple production lines and plants.

Potential Challenges and Risks

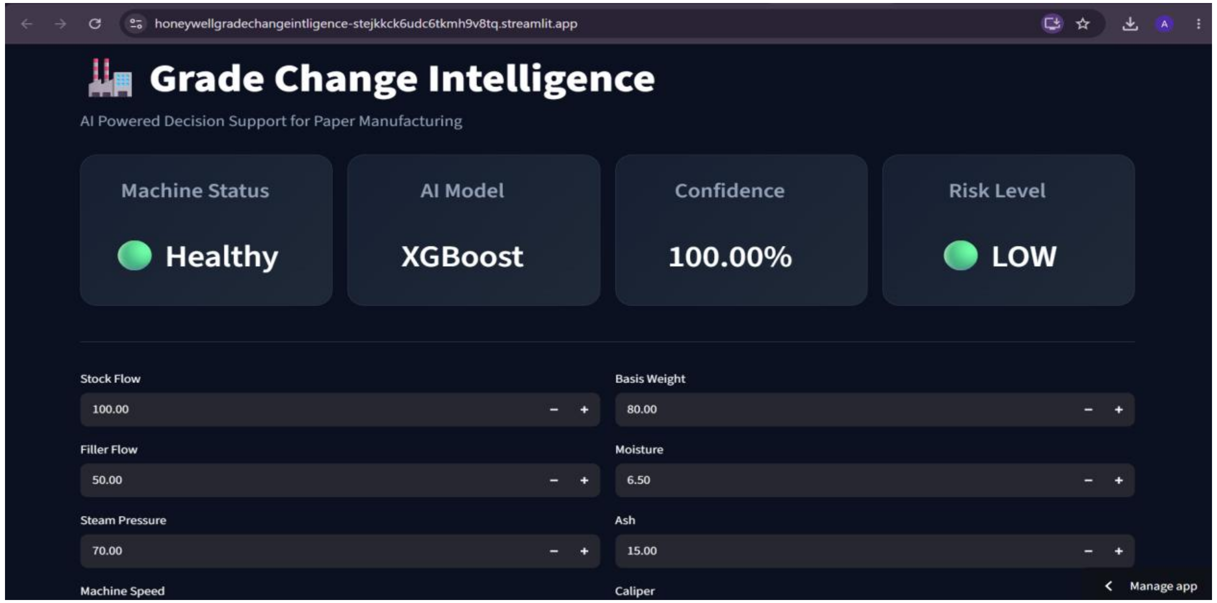
- Model accuracy depends on the quality and availability of production data.
- Process variations may require periodic model retraining and validation.
- Integration with legacy industrial systems can be technically challenging.

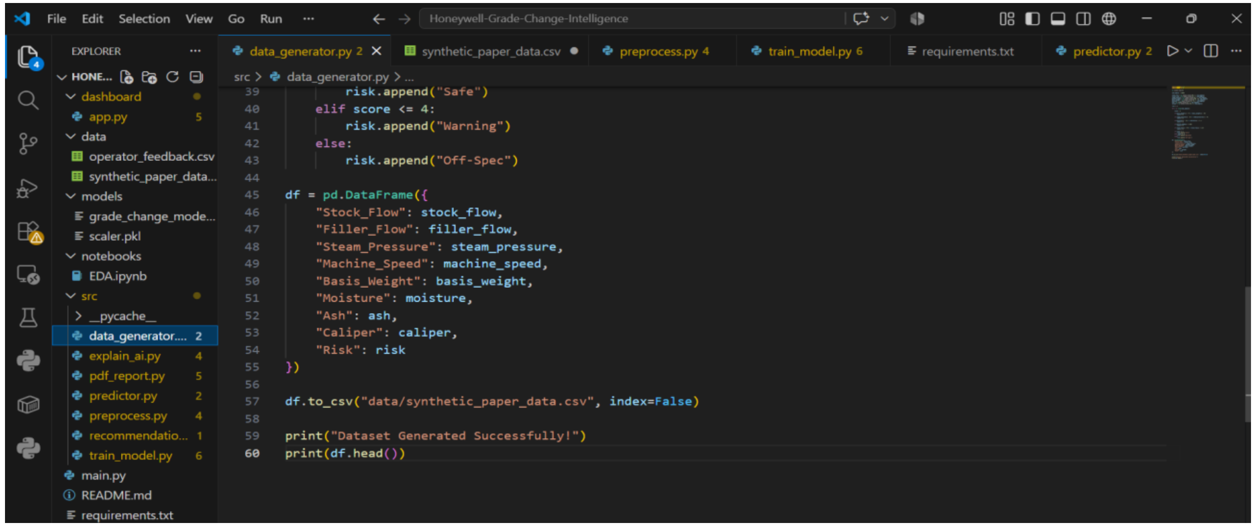
Strategies for Overcoming These Challenges

- Continuously retrain the model using new production and operator feedback data.
- Implement data validation and preprocessing to improve prediction reliability.
- Design a modular architecture for seamless integration and future scalability.

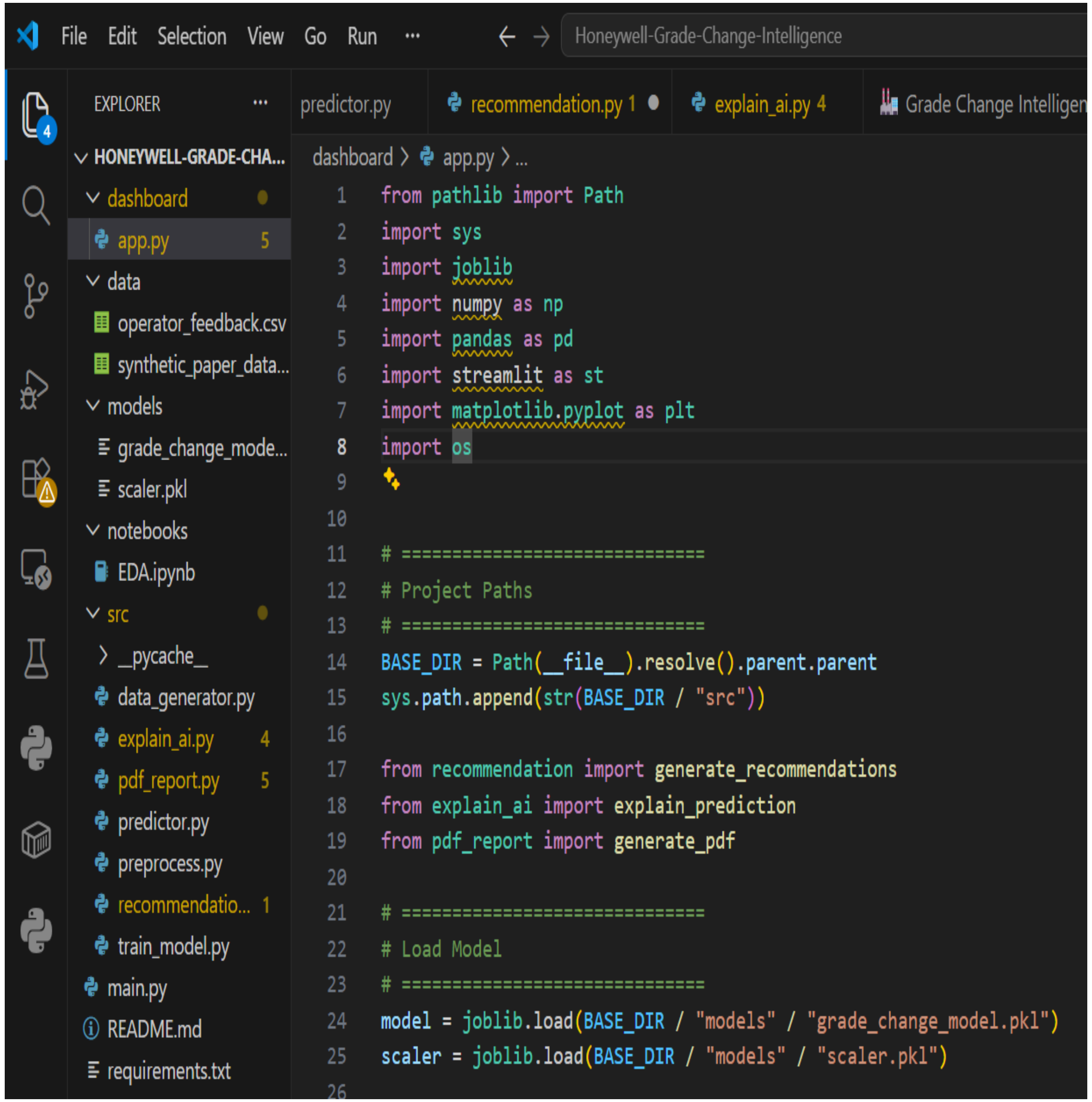
ARTIFACTS

Dashboard Snaps:





Snippet of the Machine Learning pipeline:



RESEARCH AND REFERENCES

Details / Links of the reference and research work:

- **XGBoost** – Gradient boosting machine learning framework used for accurate classification and predictive analytics.
- **SHAP (Explainable AI)** – Explainability technique that identifies the contribution of each feature to the model's prediction.
- **Scikit-learn** – Python machine learning library used for data preprocessing, model utilities, and evaluation.
- **Streamlit** – Open-source Python framework for building interactive data science and AI web applications.
- **Pandas** – Data analysis library used for data cleaning, manipulation, and structured data processing.
- **ReportLab**– Python library used to generate professional PDF reports programmatically.

Research Paper

1. Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." *Advances in neural information processing systems* 30 (2017).
2. Chen, Tianqi, and Carlos Guestrin. "Xgboost: A scalable tree boosting system." In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785-794. 2016.